EMCH 260
SOLID MECHANICS

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Combined Loadings

8

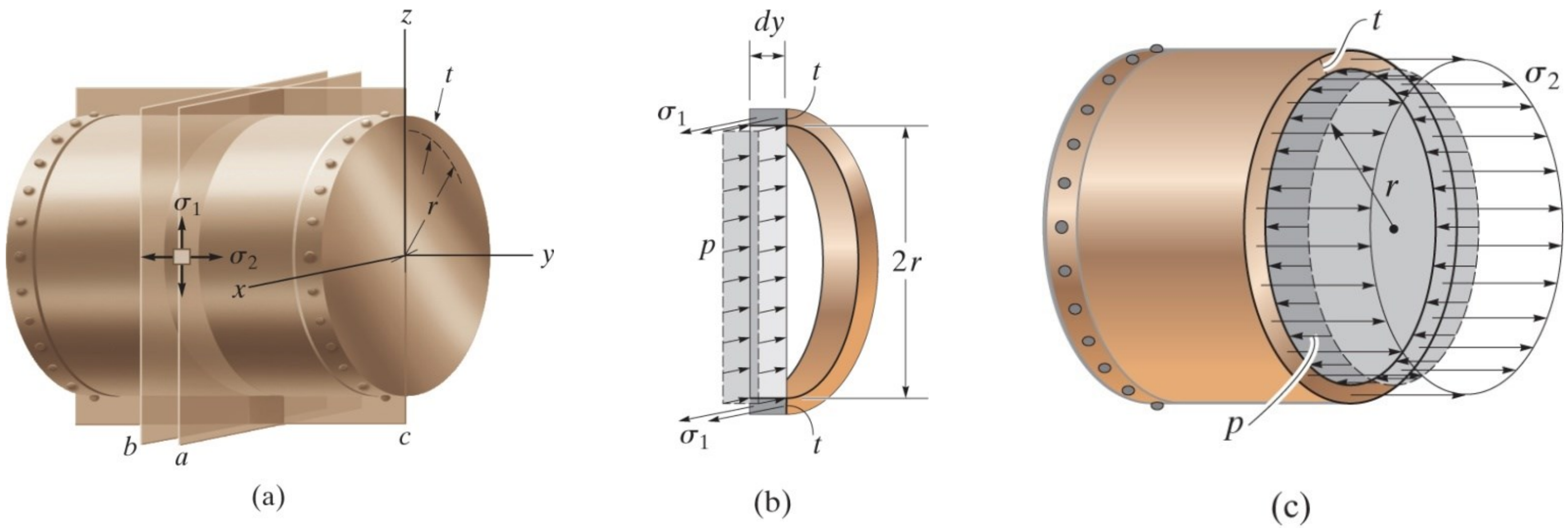
Chapter 8.2

✓ Determine stresses developed in thin-walled pressure vessels

THIN-WALLED PRESSURE VESSELS

- Cylindrical or spherical vessels are commonly used in industry to serve as boilers or tanks
- When under pressure, the material which they are made of, is subjected to a loading from all directions
- However, we can simplify the analysis provided it has a thin wall
- "Thin wall" refers to a vessel having an inner-radius-thickness ratio of 10 or more ($r/t \geq 10$)
- When $r/t = 10$, results of a thin-wall analysis will predict a stress approximately 4% less than actual maximum stress in the vessel

CYLINDRICAL VESSELS



Due to uniformity of loading, an element of the vessel is subjected to **normal** stresses σ_1 in the *circumferential or hoop direction* and σ_2 in the *longitudinal or axial direction*

$$\text{Hoop direction}: \sigma_1 = \frac{pr}{t}$$

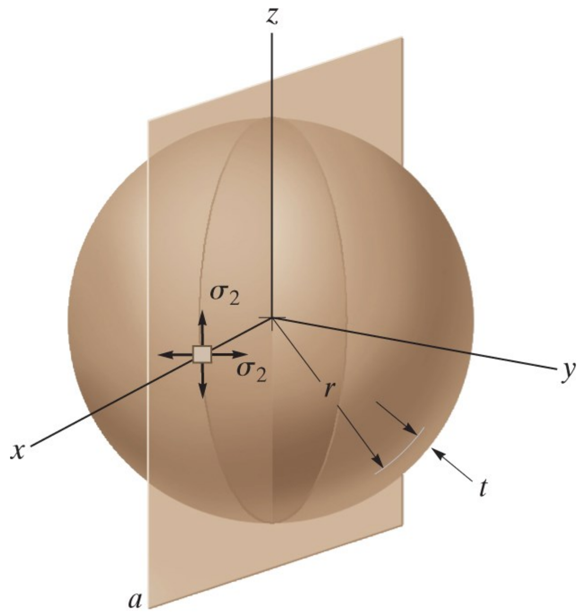
$$\text{Longitudinal stress}: \sigma_2 = \frac{pr}{2t}$$

COMPARING STRESS IN HOOP AND LONGITUDINAL DIRECTION

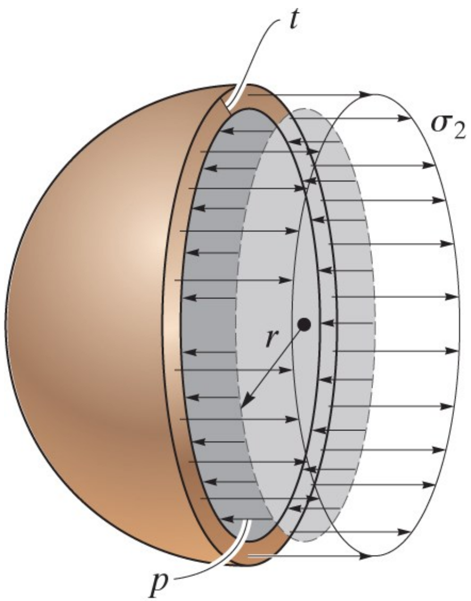
Comparing stress in hoop and longitudinal directions, note that the hoop or circumferential stress is twice as large as the longitudinal or axial stress.

The thin-walled pipe subjected to an excess gas pressure usually ruptures in the hoop direction.

SPERICAL VESSELS



(a)

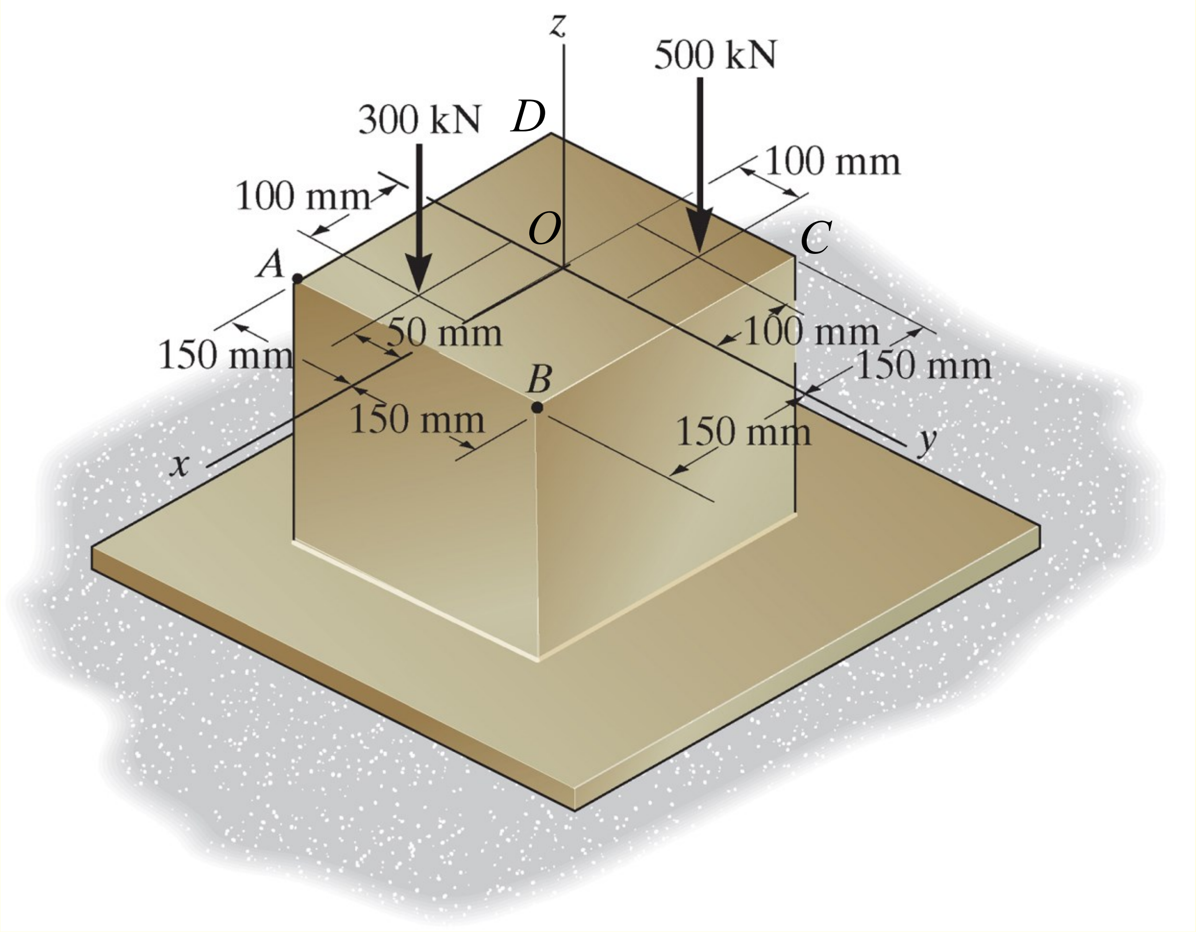


(b)

$$\sigma_2 = \frac{pr}{2t}$$

IN-CLASS EXAMPLE 2

Determine the normal stress at corners *A* and *B* of the column.

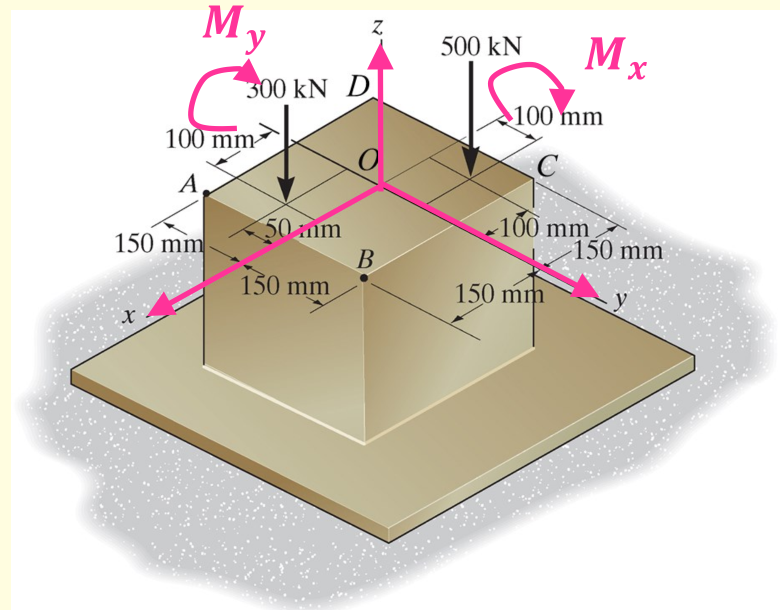


IN-CLASS EXAMPLE 2, Cont'd

Solutions

Step 1: Internal Loading

The member is already sectioned along the *ABCD* plane, where the stress is to be determined. The *xyz* coordinate system is established with origin at the centroid point *O*.



The resultant internal normal and shear force components and the bending and torsional moment components can be obtained as:

- Normal force in *z*-direction: $P = -300 - 500 = -800\ \text{kN}$

- Bending moment components:

$$M_x = 300 \times 0.05 - 500 \times 0.1 = -35\ \text{kN} \cdot \text{m}$$
$$M_y = 300 \times 0.1 - 500 \times 0.1 = -20\ \text{kN} \cdot \text{m}$$

NOTE: The sign of forces and moments is **not** so critical. However, you have to be very careful about what kind of stress these resultant forces induce (tensile or compressive stress?)

IN-CLASS EXAMPLE 2, Cont'd

Solutions, Cont'd

Step 2: Stress Components

- For normal force in *z*-direction the uniform normal-stress at A & B is

$$\sigma = \frac{P}{A} = \frac{800 \times 10^3}{0.3 \times 0.3} = 8.889 \text{ MPa}$$

(Compressive)

- For moment in *x*-direction, the stress is

$$\sigma_A = \frac{M_x c_A}{I_x} = \frac{35 \times 10^3 \times 0.15}{\frac{1}{12} \times 0.3 \times 0.3^3} = 7.778 \text{ MPa}$$

(Tensile)

$$\sigma_B = \frac{M_x c_B}{I_x} = \frac{35 \times 10^3 \times 0.15}{\frac{1}{12} \times 0.3 \times 0.3^3} = 7.778 \text{ MPa}$$

(Compressive)

IN-CLASS EXAMPLE 2, Cont'd

Solutions, Cont'd

Step 2: Stress Components, Cont'd

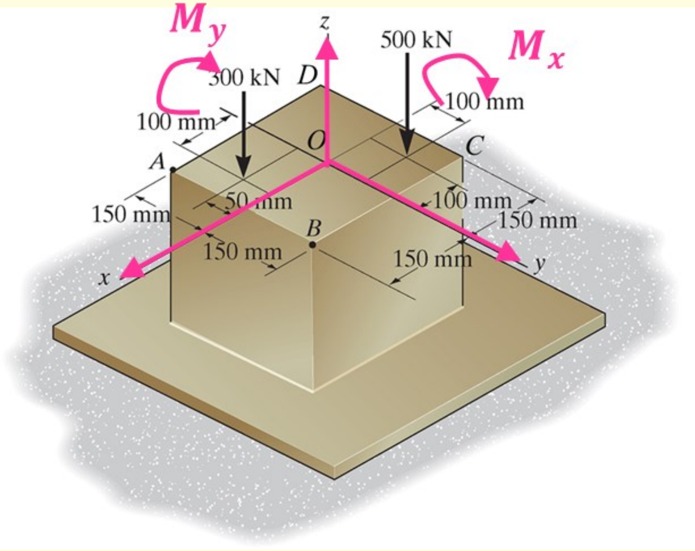
- For moment in *y*-direction, the stress is

$$\sigma_A = \frac{M_y c_A}{I_y} = \frac{20 \times 10^3 \times 0.15}{\frac{1}{12} \times 0.3 \times 0.3^3} = 4.444\ \text{MPa}$$

(Tensile)

$$\sigma_B = \frac{M_y c_B}{I_y} = \frac{20 \times 10^3 \times 0.15}{\frac{1}{12} \times 0.3 \times 0.3^3} = 4.444\ \text{MPa}$$

(Tensile)



OUT-OF-CLASS READING

- **Textbook, pp. 413 – 416**
  - **In particular, Example 8.1 (pp. 416)**